

- Q.26 Explain pot capacity.
 Q.27 What is pinch off? Explain its importance.
 Q.28 Discuss calculation for clamp pressure for compression mould.
 Q.29 Explain die head assembly for blow mould.
 Q.30 Explain neck design for injection blow mould.
 Q.31 Explain temperature and pressure control for extruder process.
 Q.32 Discuss parison programming and control.
 Q.33 Discuss principle of transfer moulding.
 Q.34 Explain Mandrel design for Injection blow mould.
 Q.35 Explain various types of cooling for Blow moulds.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Write short note on:
 a) Advantages and disadvantages of compression moulds.
 b) Die geometry, design of sprue and runners, gate used in plastic processing.
- Q.37 Write short note on :
 a) Difference between compression and Transfer moulding
 b) Draw diagram of die used in wire and cable coating.
- Q.38 Explain in detail types of mould used for Compression moulding with their advantage and limitation.

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6th Sem / Plastic Engineering, Chem Engg. (Spl Polymer Tech.)

Subject:- Design of Dies and Mould-II

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What is the name of line which lies between two mould halves?
 a) Parting Line b) Center Line
 c) Matching Line d) Vertical Line
- Q.2 What is the advantage of preheating the raw material in compression moulding?
 a) Increase the Flash b) Decrease the strength
 c) Reduce the flow
 d) Reduce moulding pressure
- Q.3 Which part is an opening at the entrance of the cavity?
 a) Runner b) Gate
 c) Sprue d) Core
- Q.4 The purpose of a fan gate is to _____
 a) Speed solidification
 b) Control injection speed
 c) Spread material over a large area
 d) Redirect the flow of material

- Q.5 It is the _____ operation that sets the shape of thermoplastics.
- Heating
 - Cutting
 - Grinding
 - Cooling
- Q.6 The usual range of mold draft angles is :
- 5° to 10° per side
 - 1° to 4° per side Bevel
 - 1° to 2° per side
 - 6° to 12° per side
- Q.7 The mold design known as 'runner-less' is the:
- hot-runner mold
 - two-plate cold-runner mold
 - insulated-runner mold
 - None of the above
- Q.8 Transfer molding process combines the principle of _____ and transfer of polymer charge.
- Injection
 - Compression
 - Blow
 - Rotation
- Q.9 A three-plate mold has:
- no sprue bushing
 - a stripper plate
 - integrated sprue pickers
 - a larger gate system
- Q.10 Draft is provided for :
- Improving strength of component
 - Easy removal of component
 - Increase stiffness
 - Proper shrinkage

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Register ring is used to align the mould with machine platen. (True/False)
- Q.12 A moulding which has a recess or projection is termed as undercut moulding. (True/False)
- Q.13 Automatic degating with fewer marks is also called a sub surface or submarine gate. (True/False)
- Q.14 _____ is round cross section and has taper along its length.
- Q.15 _____ type of cooling is usually employed to cool core of long cylindrical products.
- Q.16 A good runner should convey melt with minimum loss of pressure. (True/False)
- Q.17 How many plates in the mould for transfer moulding process (three or four)?
- Q.18 _____ and _____ are the main elements of mould.
- Q.19 Name the two materials used for blow moulding process.
- Q.20 Give two functions of band heaters.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain various materials used in blow mould.
- Q.22 Discuss various types of dies used in extrusion.
- Q.23 Explain Die swell and its importance.
- Q.24 Give advantages of plunger transfer mould.
- Q.25 State limitations of compression moulds.

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